

AEYAN ASHRAF

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EDUCATION

University of Massachusetts Amherst

Master of Science in Computer Science - GPA: 3.83/4.0

📅 Sep 2023 – May 2025

📍 Amherst, MA

Aligarh Muslim University

Bachelor of Technology (B.Tech.) in Computer Engineering- GPA: 9.05/10

📅 Aug 2019 – May 2023

📍 Aligarh, India

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL (MySQL), Bash, C++.

Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, TensorFlow Extended (TFX), MLflow, Hugging Face.

Cloud / MLOps: Docker, Kubernetes, Airflow, GitHub Actions, Google Cloud Platform (GCP).

Tools & Technologies: Git, Firebase, Flutter, Linux, L^AT_EX, Large Language Models.

WORK EXPERIENCE

Monash University

📅 Sep 2025 – Present

Research Assistant – LiteCrackSeg (under Dr. Asad Malik)

📍 Remote

- Co-developed LiteCrackSeg, a 0.55M-parameter, 3.40 GFLOPs hybrid CNN–Transformer model for efficient crack segmentation on edge devices, integrating depthwise separable convolutions with LoRA-enhanced MobileViT blocks.
- Designed the Dynamic Edge Enhancement Module (DEEM), which fuses frozen Sobel and Laplacian-of-Gaussian filters with dynamic convolutions to refine thin and fragmented crack boundaries across diverse backgrounds.
- Proposed Context-Aware Domain Adaptive Regularization (CoDAR) to penalize overconfident encoder channels, improving domain robustness and zero-shot cross-dataset generalization without inference-time overhead.
- Validated the model on Crack500, DeepCrack, and TUT datasets, achieving state-of-the-art F1 and mIoU while maintaining real-time performance on NVIDIA Jetson Orin Nano (64 FPS, low energy per inference).

Goldman Sachs

📅 Jan 2025 – Aug 2025

Machine Learning Researcher

📍 Amherst, MA

- Developed a scalable RAG pipeline for SEC 10-K filings using Python, LangChain, and GPT OSS 120B, automating data ingestion, chunking, and multi-hop QA generation across enterprise-scale corpora.
- Engineered an end-to-end ML pipeline for factoid extraction and answer grounding; implemented iterative LLM-based refinement achieving 94% precision in automatic evaluations.
- Designed and deployed quality control workflows using LLM-as-a-judge scoring for fairness, reasoning traceability, and numeric fidelity across 1,000+ QA samples.
- Benchmarked AI systems using precision, recall, nDCG, BLEU, and ROUGE-L to validate retrieval/generation robustness in high-stakes document QA systems.

PROJECTS

PerfectPitch — Next.js, Tailwind CSS, Firebase, Gemini.ai, LLM, Assembly.ai

[GitHub](#)

- Built an LLM-integrated mock interview and resume feedback platform using real-time voice-to-text (Assembly.ai) and Gemini-generated coaching.
- Implemented NLP pipelines for ATS optimization, improving job match scores and interview preparedness through structured feedback and keyword analysis.

A Vision Transformer-Based Approach to Bearing Fault Classification via Vibration Signals — Python, PyTorch, ViT, Time-Frequency Analysis [Paper](#)

- Transformed 1D vibration signals into 2D time–frequency representations and trained a Vision Transformer-based model for fine-grained bearing fault classification.
- Achieved 98.8% accuracy across 13 bearing defect classes, demonstrating the effectiveness of ViT-style global attention for rotating machinery health monitoring.
- Analyzed model robustness to noise and domain variations, informing design choices for reliable, data-driven predictive maintenance pipelines.

SSORT: Semantic Segmentation for Off-Road Traversability — (Bachelor Thesis) Python, HuggingFace, PyTorch, Flask

[GitHub](#)

- Enhanced semantic segmentation for off-road terrain using SegFormer, improving IoU and pixel accuracy by 10% over traditional encoder–decoder models.
- Deployed a scalable inference API with Flask and benchmarked performance (F1 score, pixel accuracy) on geospatial datasets for robust evaluation.

SELECTED PUBLICATIONS

- **LiteCrackSeg: A Hybrid Architecture for Efficient Crack Segmentation on Edge Devices.** (Submitted to CVPR 2026).
- **FINHOP:** Benchmarking Retrieval-Augmented Generation for Multi-Hop Questions on Long Financial Documents. (Manuscript under preparation).
- **A Vision Transformer-Based Approach to Bearing Fault Classification via Vibration Signals.** 14th Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC).

AWARDS & CERTIFICATIONS

- Hackathons (Rank): Hack413 (1st place), HackUMass XI (1st place).
- Awarded the **Sir Syed Mentoring and Assistance Fellowship (SSMAF)** recognizing academic excellence and research potential.